UNIVERSITY OF WOLLONGONG AUSTRALIA

# Tutorial 04

## Time-response evidence and model identification

Preparation: Lecture 04: first-order steps and time constants; second-order transients, identification, pole locations and model limits.
Submission credit: 1% of the course grade for submitted work; correctness is not marked.
In class: complete the basic calculations and response readings in Q1–Q3. Extensions A and B are optional; the damping-formula derivation and model reduction are not core tasks.

## Question 1 Quick checks on time response

Give a brief reason or calculation.
a) Thermometer step. A unity-gain first-order thermometer has time constant 2 s. It moves from a 20°C bath to a 30°C bath at t = 0. Using deviations from 20°C and zero initial deviation, write its transfer function and temperature response.

b) Time constant and ramp error. For part (a), give the final temperature and the percentage of the change completed at t = 2 s. Estimate the 2% settling time using 4τ. If the water later warms steadily at 1°C/s, use $e_{ss}=B\tau$ to find the steady reading error.

c) Damping regimes. For $G(s)=9/(s^2+3s+9)$, identify $\omega_n$, ζ and the damping type. Classify standard second-order systems with ζ = 0, 1 and 1.5. Which has sustained oscillation and which have no oscillatory overshoot?

d) One pole pair. A standard constant-numerator second-order system has poles $-7\pm j7\ s^{-1}$ and damping ratio ζ = 0.707. Use $t_s\approx 4/[-\mathrm{Re}(p)]$ to estimate its 2% settling time. Does this estimate meet $t_s\le 0.60$ s, and is ζ ≥ 0.6?

## Question 2 Identify a test rig from a force-step experiment

A fixed-base test rig is modelled by $mx'' + bx' + kx = F$ with a constant force-to-displacement numerator. A 4 N force step is applied from rest. The measured final displacement is 0.0200 m, first peak is 0.02326 m, and first-peak time is 0.800 s. Treat the peak data as rounded measurements.

For the main calculation, take $\zeta = 0.5000$ as given.
a) Find stiffness, overshoot ratio and damped frequency.

b) Recover natural frequency, mass and damping coefficient. Write the identified transfer function with units.

## Question 3 Read and interpret transient responses

Both normalized systems start from rest and have unit DC gain and damping ratio ζ = 0.5. Only natural frequency changes: $\omega_n$ =4 rad/s for A and 8 rad/s for B. The curves are calculated standard second-order responses.

$G_A(s)=16/(s^2+4s+16)$, $G_B(s)=64/(s^2+8s+64)$

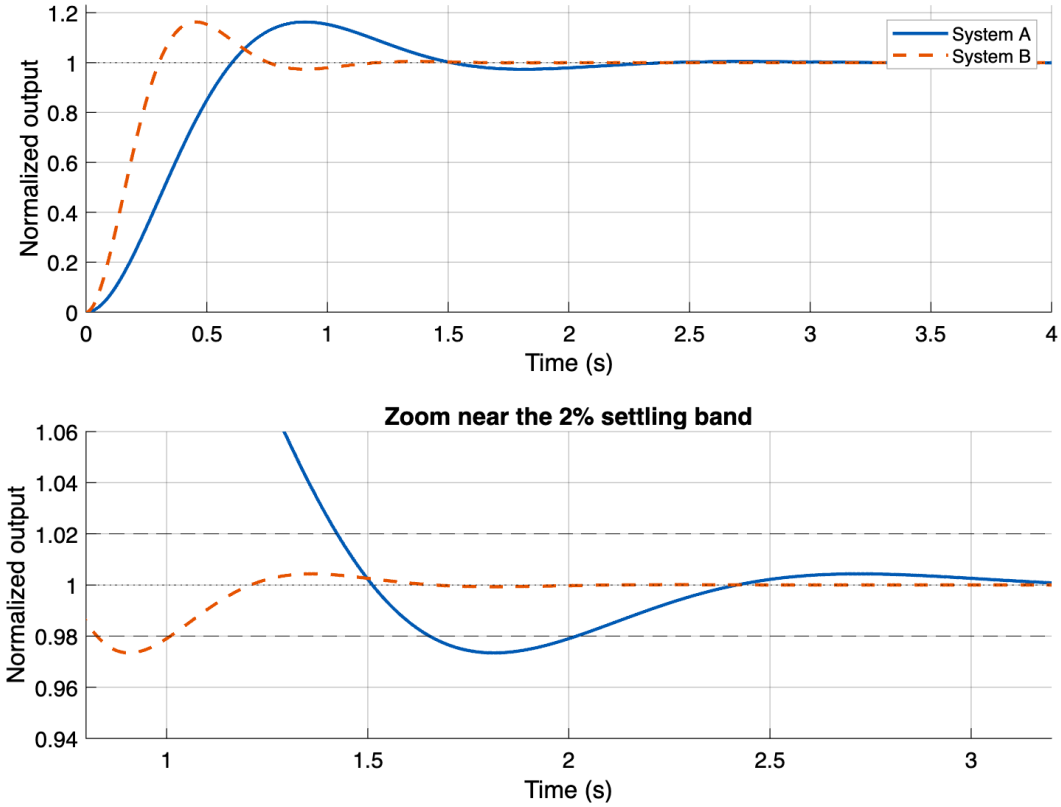



**Figure 1.** Unit-step responses and a zoom near the ±2% settling band.

a) Read System A's final value, 0–100% rise time (first arrival at the final value), first-peak time and peak height. Calculate percentage overshoot. Read the 2% settling time from the lower panel: use the last crossing into the band, not the first entry.

b) Compare B with A: does doubling natural frequency change percentage overshoot or final value? Which satisfies both overshoot ≤ 20% and 2% settling time ≤ 1.2 s? Can these normalized force-step curves alone certify the seat project's acceleration and relative-travel limits under a floor bump?

c) For a physical force-input plant, keep the other parameters and force-step height fixed. Increasing b within the underdamped range: what happens to overshoot? Increasing m: does the final displacement change? Increasing k: must the decay envelope become faster? Use $x(\infty)=F_0/k$ and decay rate $\sigma=b/(2m)$.

MATLAB verification. Simulate the unit-step responses of A and B from rest on one time vector.

Submit one figure showing both responses and a second panel of deviations from the final value, with the ±2% band. Use contrasting colours and line styles. Report rise time, peak time, overshoot and last-entry settling time; compare with **Figure 1** and the two requirements. Base MATLAB is sufficient.

## Extension A Identify damping from the vibrating steel ruler

After class / optional laboratory connection — brief introduction in class
Reference practical: Model Identification from Experimental Data. Clamp a steel ruler-like cantilever, deflect its tip by 10–20 mm and release from rest. Two strain gauges in a Wheatstone bridge, DAQ and LabVIEW record the decay for 5–10 s; zero the strain before release and record three runs.
Use Reference/Cantilever_beam.m to convert strain to tip deflection and select a decay interval. Set the data filename and verify the sampling interval (the script assumes 0.001 s).
For the reference beam, L = 230 mm, h = 1 mm, and x = 32 mm is the strain-gauge position measured from the clamp. Here z(t) is tip deflection relative to equilibrium and ε is dimensionless strain. The conversion is:

$$z(t) = 2L^3 \varepsilon(t) / [3h(L - x)]$$

Keep all lengths in the same units; using mm gives z in mm. Convert microstrain to dimensionless strain first. Remove the equilibrium offset before reading peak amplitudes.
Worked-example readings below are illustrative, not an actual LabVIEW record: selected positive peaks $z_0$ = 15.0 mm and $z_6$ = 7.50 mm are separated by $t_6 - t_0$ = 0.3774 s. There are six complete periods between them. For the optional model calculation, use effective mass $M_e$ = 0.01277 kg and effective stiffness $k_e$ = 127.4 N/m from the reference beam geometry.

a) Describe the route from the measured strain to the deflection plot. What does the decreasing peak amplitude tell you? Is this a force-step experiment?

b) Use the six-period interval to calculate average damped period $\tau_d$, damped natural frequency $\omega_d$, logarithmic decrement δ, damping ratio ζ and natural frequency $\omega_n$. Use the formulas below.

$$\tau_d = (t_6 - t_0)/6, \quad \omega_d = 2\pi/\tau_d, \quad \delta = (1/6)\ln(z_0/z_6)$$

$$\zeta = \delta/\sqrt{(4\pi^2 + \delta^2)}, \quad \omega_n = \omega_d/\sqrt{(1 - \zeta^2)}$$

c) Optional follow-through: use $b_e = 2\zeta\sqrt{(k_e M_e)}$ to estimate effective damping. Write the force-to-tip-deflection transfer function. Explain why its zero-initial-state transfer function alone does not generate the release-test response.
Extension source: Reference/Practical - Model Identification.docx, Experimental Procedure and Data Analysis Steps 1–2, 3, 5, 6–8; Reference/Cantilever_beam.m. Six-period averaging and symbols follow the practical; illustrative readings are supplied here for calculation.

## Extension B Damping inference and a reduced model

Optional strengthening questions. Neither part is required for Q2 or Q3.
a) For the Q2 force-step data, the overshoot ratio is 0.163. Starting with the supplied overshoot relationship, derive the inverse formula for ζ and evaluate it.

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$$

b) A separate model is $G_H(s)=100/[(s+10)(s^2+2s+10)]$, with poles −10 and −1±j3 $s^{-1}$. Identify the slower mode and approximate s+10 by 10 to obtain a second-order model.

## Method reference

Quick-check sources: Reference/tutorials/Tute 5 Tutorial 5 Questions, Q2 (thermal first-order model); Reference/tutorials/Tute 6 Tutorial 6 Questions, Q1–Q2 (damping and poles). Ogata, Modern Control Engineering, 5th ed., Secs. 5–2 and 5–3, pp. 161–177. Q1 uses adapted data and the previous short tasks.
Method source: Ogata, Modern Control Engineering, 5th ed., Problem A-5-5, pp. 236–237 (physical parameter identification); System Dynamics, 4th ed., Chapter 8 (transient response). Q3 uses Lecture 4 standard response measures, parameter trends and dominant-mode reasoning with independently selected models. Page numbers refer to printed pages.
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